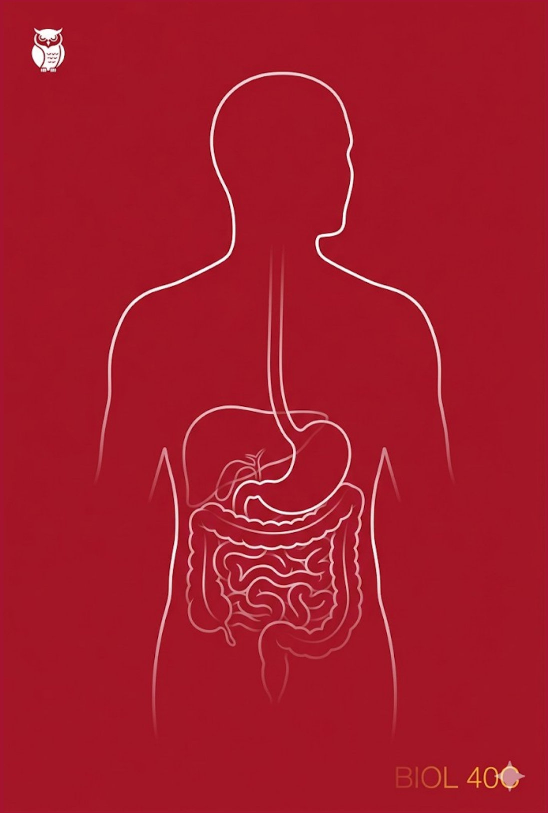
BIOL 40C

Human Anatomy & Physiology III

Spring 2026 | Foothill College

Giorgio Lagna, Ph.D. | Monday, April 6, 2026



Meet Your Instructor

Giorgio Lagna

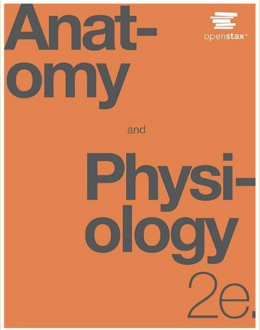
(call me JOR-joe or Dr. LAN-ya)

- M.S. Biology, University of Rome
- Ph.D. Molecular Biology, Rockefeller University, NYC
- Research: Tufts University, UCSF Cardiovascular Research Institute
- Industry: Biotech start-ups in molecular & cellular biology
- Teaching: USF, UC Berkeley Extension, Santa Clara University, Foothill College

> *"I'm passionate about making learning engaging, accessible, and interactive."*

Course Materials & Resources

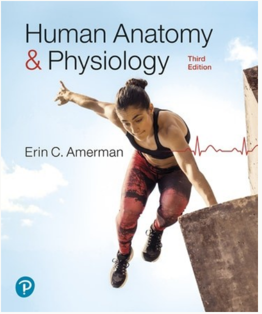
REQUIRED



OpenStax Anatomy and Physiology 2e

FREE online at openstax.org

RECOMMENDED



E.C. Amerman, Human Anatomy & Physiology, 3rd ed.

Pearson e-text ~$9.99/month

All course resources are on Canvas. I'll point you to everything you need.

How This Course Works

Lectures: MW 10:00–11:50 AM, Room 8403

Active, case-based, lots of group work (Giorgio Lagna)

Labs: M or W 12:30–3:20 PM, Room 8710

Hands-on anatomy, dissection, observation (Jeff Schinske)

Canvas: Assignments, quizzes, readings, communication

Your hub for everything between classes

This is an in-person course. Showing up and participating is how you learn.

Your Weekly Study Rhythm

PREPARE

- Read textbook chapter
- Complete Active Learning Workbook (ALW)
- Take Key Concepts Quiz (2 attempts, keep highest)

Due: Sunday 11:59 PM

→

LEARN

- Attend lectures (MW)
- Polls, group work, clinical cases
- We build on your preparation

In class

→

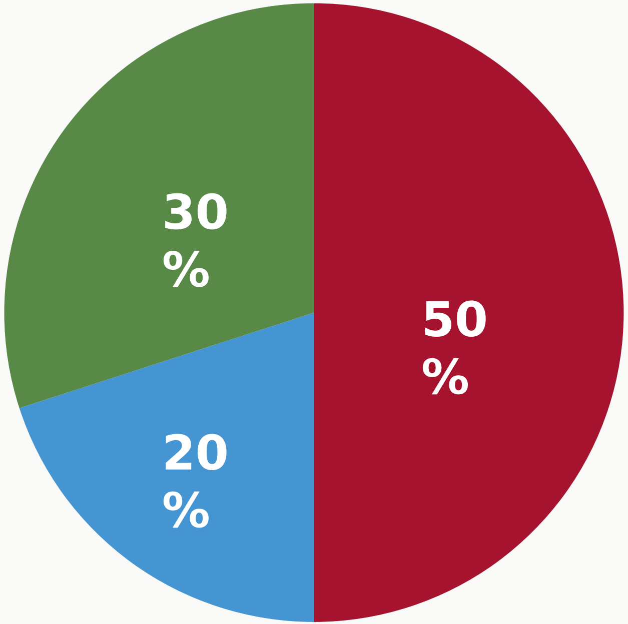
DEMONSTRATE

- Running Case Study Quiz
- Application Activities

Due: Saturday 11:59 PM

The ALW and KC Quiz prepare you BEFORE lecture. The RCS Quiz tests you AFTER.

How Your Grade Works



- **Exams: 50%**
  Midterm (May 18) + Final (June 27). Case-based, reasoning over recall.
- **Homework & In-Class: 20%**
  ALW, KC Quizzes, RCS Quizzes, Application Activities, Polls
- **Laboratory: 30%**
  Coordinated by Prof. Schinske. Hands-on, essential.

Detailed breakdown is in the syllabus on Canvas.

Share with Your Group of Three

ACTIVITY

1. Your name
2. Why are you taking BIOL 40C?
3. Something important to know about who you are and what you value

   (e.g., cultural background, pronouns, family, pets, hobbies)

Groups of 3 with your nearest neighbors. You have 8 minutes.

Each group will share one interesting thing they learned about a groupmate.

Your Quarter at a Glance

BIOL 40C

Wk 0-1	Wk 2	Wk 3	Wk 4	Wk 5	Wk 6	Wk 7	Wk 8	Wk 9	Wk 10
	⚡	?	⚖	?		?	?	?	?
Digestive	Metabolism	Urinary	Fluid Balance	MIDTERM	Immune	Endocrine	Repro-ductive	Develop-ment	FINAL

These systems talk to each other. By Week 9, you'll see how everything connects.

Quick Poll

CHECK

Which of these systems are you MOST curious about?

Digestive	Metabolism	Urinary	Fluid Balance
Immune	Endocrine	Reproductive	Development

Raise your hand or use Poll Everywhere →

Which of these systems are you MOST curious about?

0

Digestive	0%
Metabolism	0%
Urinary	0%
Fluid Balance	0%
Immune	0%
Endocrine	0%
Reproductive	0%
Development	0%

Think-Pair-Share

ACTIVITY

"What are some things that might happen to your body if you start a new routine of vigorous exercise?"

1	2	3
THINK	**PAIR**	**SHARE**
Write down 2–3 ideas (1 min)	Share with a neighbor (2 min)	We'll discuss as a class

Try to go beyond "sore muscles." What is happening at the cellular level?

Our Goal: A Molecular & Cellular Perspective

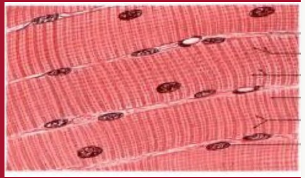
CONCEPT

GENERAL PUBLIC

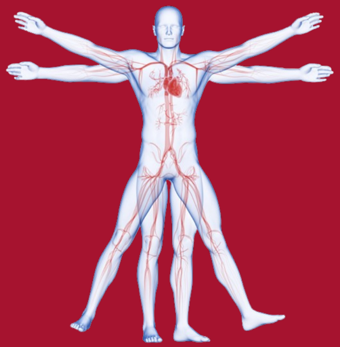
"I started exercising and my muscles are sore."

A&P SCIENTIST

"Vigorous exercise caused micro-tears in myofilaments, triggering an inflammatory response with satellite cell activation and eventual muscle fiber hypertrophy."



In this course, we always ask WHY at the cellular and molecular level. That is our lens for health and disease.



Develop a Molecular and Cellular Perspective
of the Body in Health & Disease

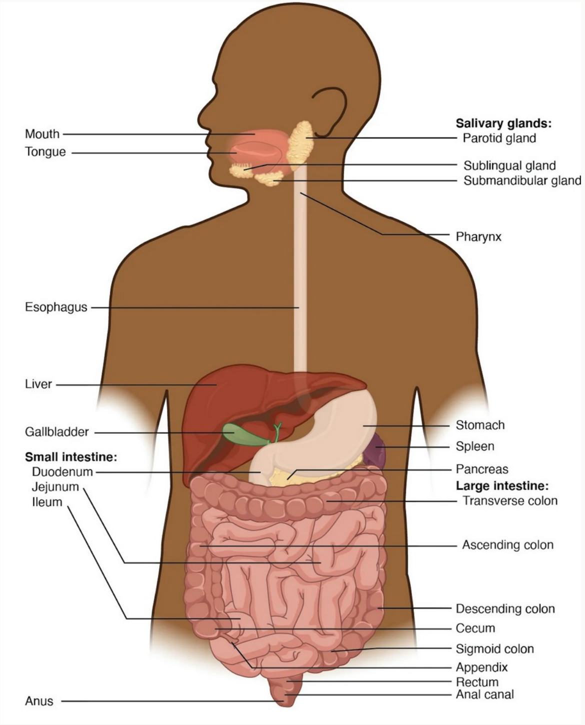
This is the thread that connects every system we study this quarter.

Let's Begin: The Digestive System

CONCEPT

Weeks 0 & 1

- You use this system every day, multiple times a day.
- It's where molecules from the outside world enter your body.
- Dysfunction here affects every other system we'll study.



By the End of Weeks 0 & 1, You Will Be Able To:

- ✓ Identify the anatomical structures of the digestive system and explain their functions
- ✓ Describe the mechanical and chemical processes that break down food
- ✓ Explain how nutrients are absorbed and waste is eliminated
- ✓ Apply your understanding to real cases of digestive dysfunction

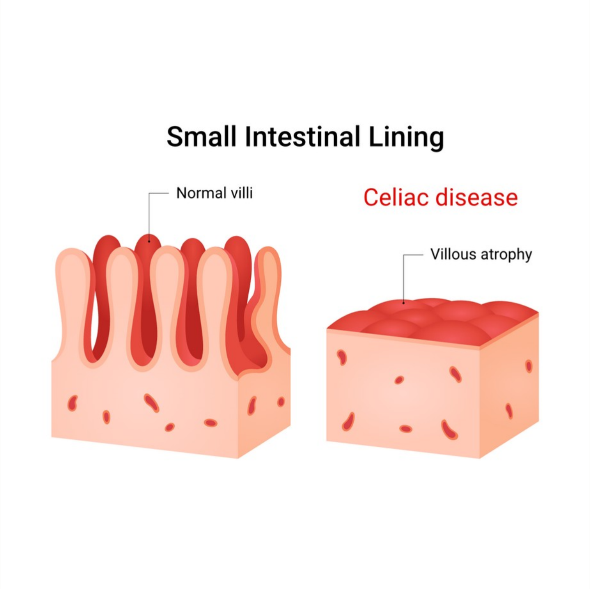
The goal is not memorization. It's learning how to reason through biological problems.

Clinical Anchor: Celiac Disease

CONCEPT

- Affects ~1% of the population
- Symptoms: abdominal pain, bloating, chronic diarrhea, malnutrition
- **Cause: immune reaction to gluten damages the lining of the small intestine**
- Consequence: can lead to anemia, osteoporosis, infertility, neurological problems

> *We'll come back to celiac disease in Class 3, after you understand intestinal structure.*



Prediction Challenge

ACTIVITY

"If the lining of your small intestine is damaged and flattened, what functions do you think would be impaired?"

Write down TWO predictions on a piece of paper.

We won't answer this today. Hold onto your predictions.

We'll revisit them in Class 3.

A Key Insight

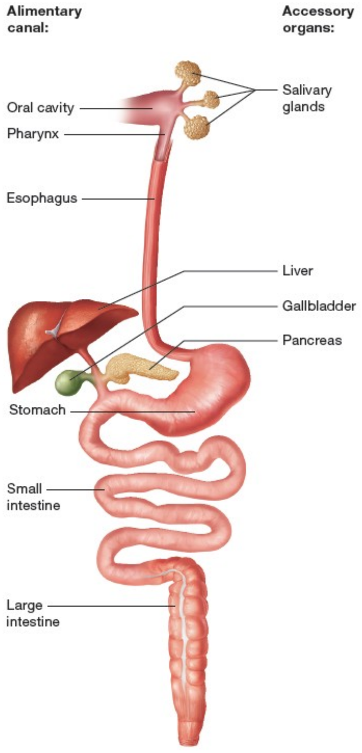
CONCEPT

Food in the digestive tract is technically

OUTSIDE your body.

Materials don't enter the body until molecules are absorbed from the tract into the blood.

The GI tract lumen is continuous with the external environment — a tube open at both ends.



(b) Schematic diagram of the digestive system

Digestive Organs: Two Categories

CONCEPT

Alimentary Canal

Mouth → Pharynx → Esophagus →

Stomach → Small Intestine →

Large Intestine

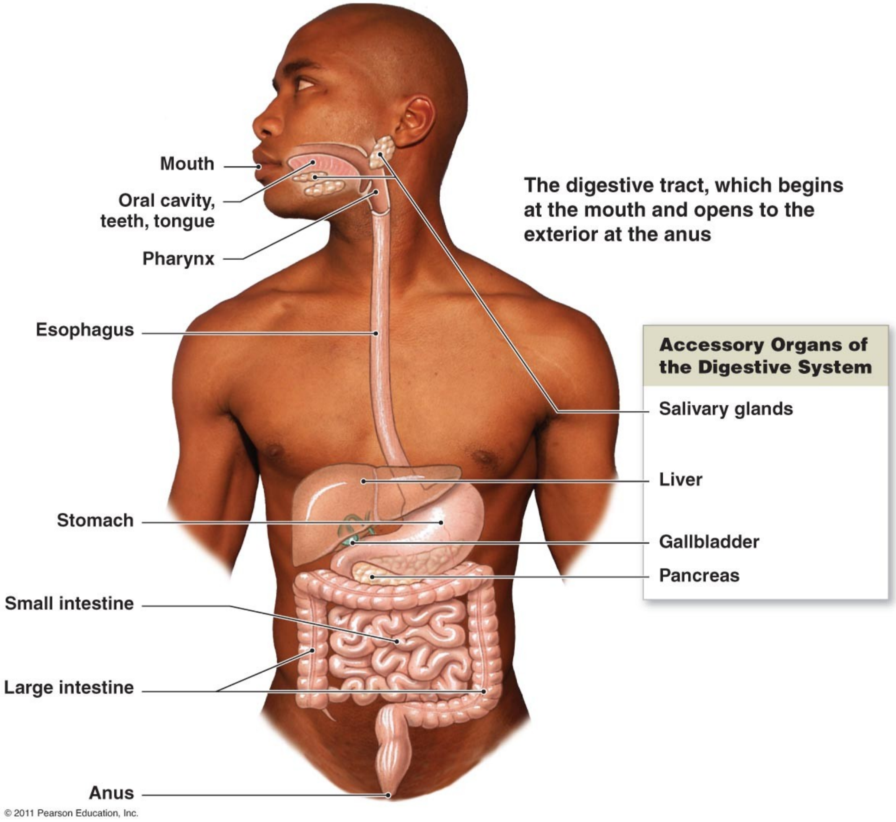
"The continuous tube from mouth to anus"

Accessory Organs

Teeth, Tongue, Salivary Glands

Liver, Gallbladder, Pancreas

"They assist digestion but food does not pass through them"



The Four Basic Digestive Processes

CONCEPT

1

INGESTION

Taking food into the mouth

2

DIGESTION

Mechanical (physical) + Chemical (enzymes)

3

ABSORPTION

Nutrients cross intestinal wall into blood/lymph

4

DEFECATION

Indigestible materials exit as feces

Every region of the GI tract specializes in one or more of these processes.

Quick Draw Challenge

ACTIVITY

1. On a blank sheet of paper, sketch the digestive tract from mouth to anus.
2. **Label as many organs as you can FROM MEMORY.**
3. You have 2 minutes.
4. Then compare with the textbook figure. Where were your gaps?

Digestion: An Adaptable, All-Terrain System

CONCEPT

- Humans thrive on incredibly diverse diets, from raw roots to double cheeseburgers.
- The GI tract seamlessly switches modes: mechanical digestion, acid, enzymes.
- **There is no single 'ancestral' diet. Our system evolved to handle variety.**

BREAK

10 minutes — we resume at 11:25

Stretch, grab water, check your phone. Be back on time.

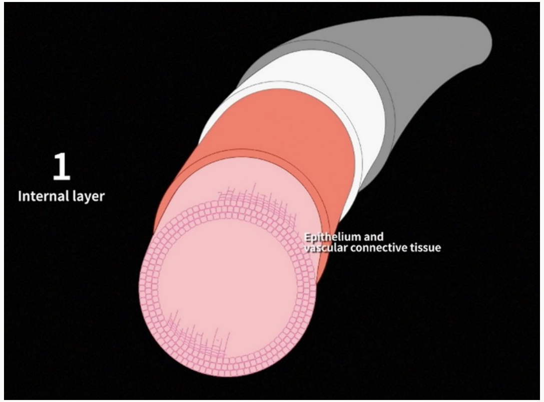
Refresher from 40A: Epithelial Tissue Types

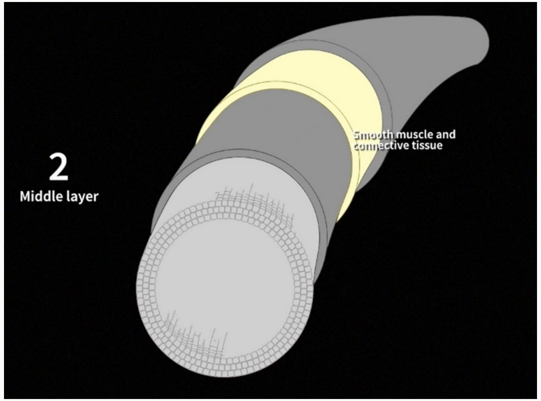
A&P REVIEW

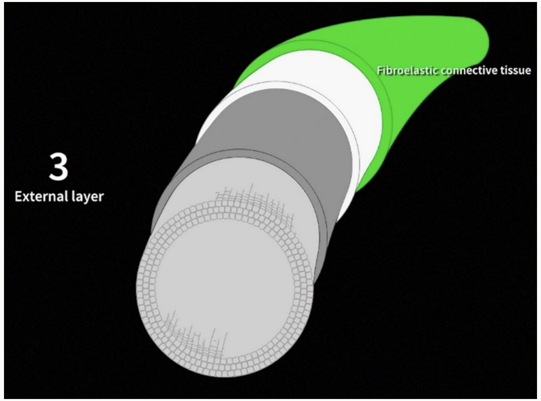
> **Before we look inside the wall of the GI tract, let's review a concept you learned in 40A.**

All tubes in the body follow a similar layered pattern.

But the type of epithelium lining the tube tells you what that organ DOES.





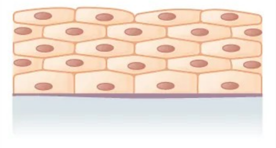
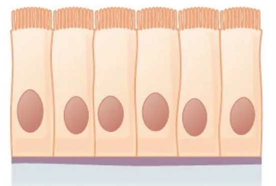
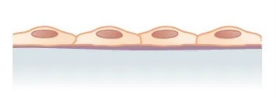


The Tissue Challenge

ACTIVITY

Discuss: Which type lines the esophagus? Which lines the stomach? Why different?

Match the tissue type to its location and function:

Tissue Type	Where?	Structure	Function
Stratified Squamous	Esophagus, skin, vagina		
Simple Columnar	Stomach, intestines		
Simple Squamous	Alveoli, capillaries, Bowman's capsule		

Histology of the Alimentary Canal: Four Layers

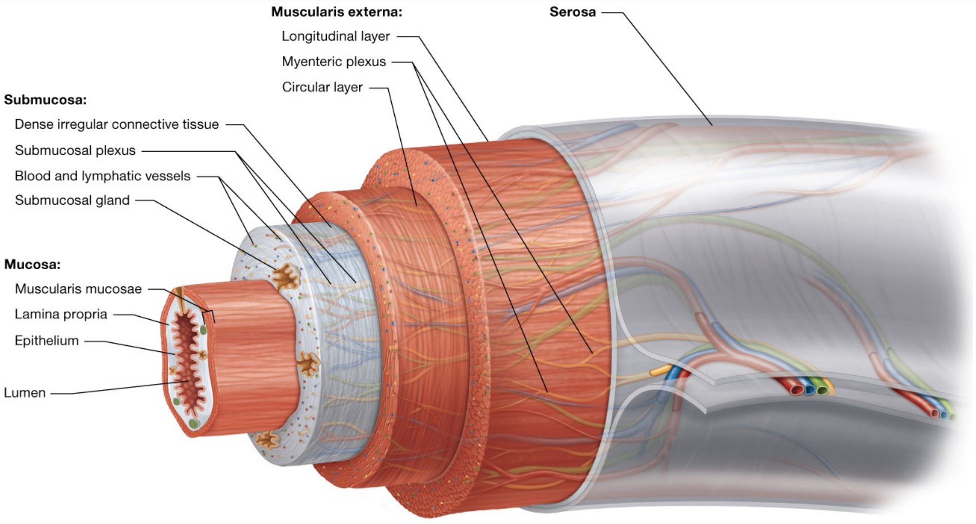
CONCEPT

1. MUCOSA

Epithelium + lamina propria + muscularis mucosae

2. SUBMUCOSA

CT, blood vessels, submucosal plexus



3. MUSCULARIS EXT.

Inner circular + outer longitudinal, myenteric plexus

4. SEROSA

Visceral peritoneum

Layer 1: The Mucosa

CONCEPT

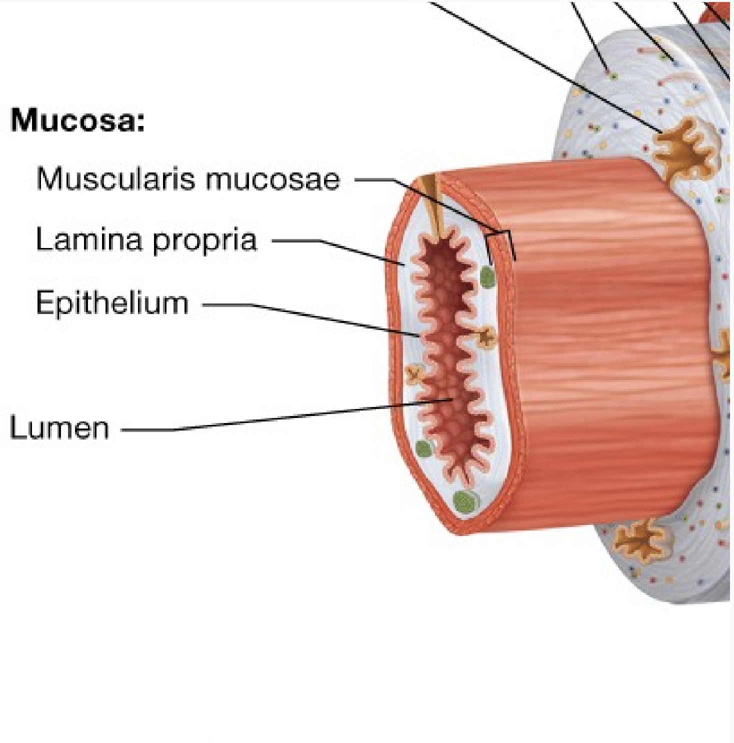
Faces the lumen (the "inside" of the tube)

Three sublayers:

- Epithelium (type varies by organ)
- Lamina propria (loose CT with blood/lymph vessels and MALT)
- Muscularis mucosae (two thin smooth muscle layers for local movement)

High mitotic rate: cells are constantly being replaced

Houses MALT (mucosa-associated lymphoid tissue) for immune defense



The Outer Three Layers

BIOL 40C

CONCEPT

SUBMUCOSA
The supply and regulation layer

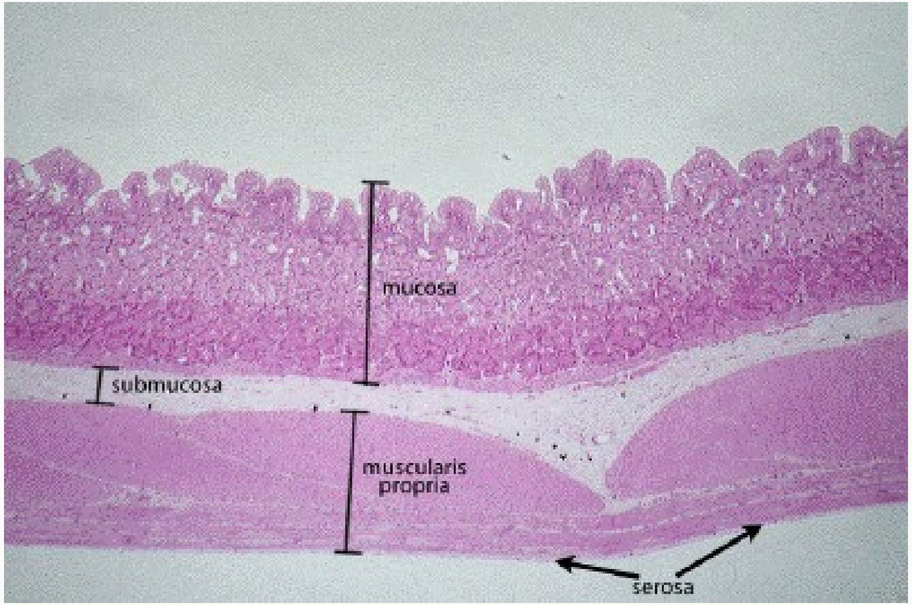
Dense irregular CT. Blood and lymphatic vessels, glands.
Submucosal plexus (nerve cluster of the enteric nervous system)

MUSCULARIS EXTERNA
The movement layer

Inner circular + outer longitudinal smooth muscle.
Myenteric plexus between layers. Controls peristalsis and segmentation.

SEROSA / ADVENTITIA
The protective outer wrapping

Serosa (intraperitoneal): simple squamous epithelium + loose CT.
Adventitia (retroperitoneal): dense irregular CT.



Your Gut Has Its Own Brain

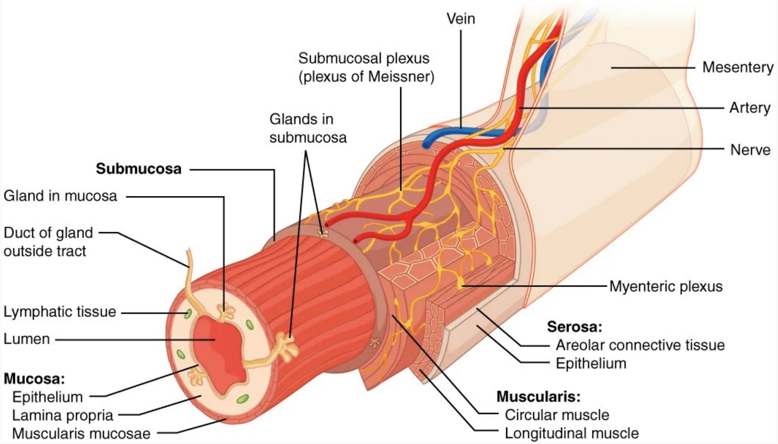
CONCEPT

The enteric nervous system (ENS) contains ~100 million neurons

It can operate INDEPENDENTLY of the central nervous system

Two plexuses:

- Submucosal plexus → controls secretion
- Myenteric plexus → controls motility



We'll see the ENS in action when we study gastric acid regulation on Wednesday.

Concept Check

CHECK

Which layer of the alimentary canal contains the nerve plexus that controls motility (peristalsis)?

A. Mucosa

B. Submucosa

C. Muscularis externa

D. Serosa

Think individually, then discuss with your neighbor.

Which layer of the alimentary canal contains the nerve plexus that controls motility (peristalsis)?

0

Mucosa

0%

Submucosa

0%

Muscularis externa

0%

Serosa

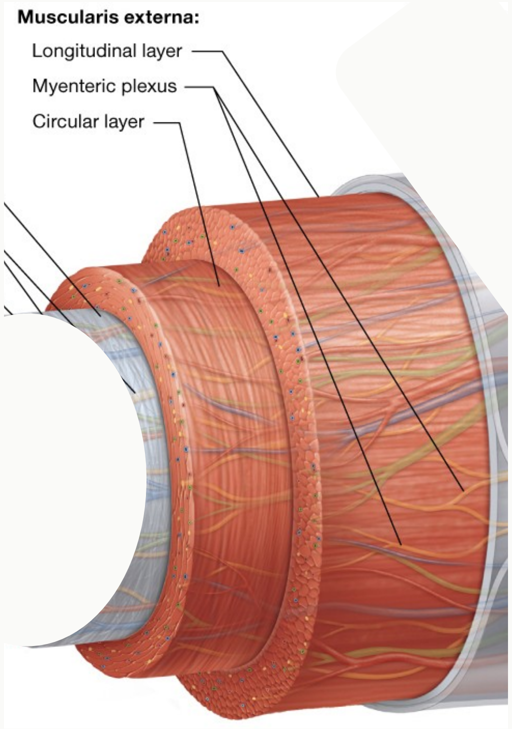
0%

Answer: C. Muscularis Externa

✓ C. Muscularis Externa

The myenteric plexus sits between the inner circular and outer longitudinal smooth muscle layers of the muscularis externa. It regulates the rate and strength of contractions (peristalsis and segmentation).

The submucosal plexus (in the submucosa) controls secretion and blood flow, not motility.



The Peritoneum: The Body's Largest Serous Membrane

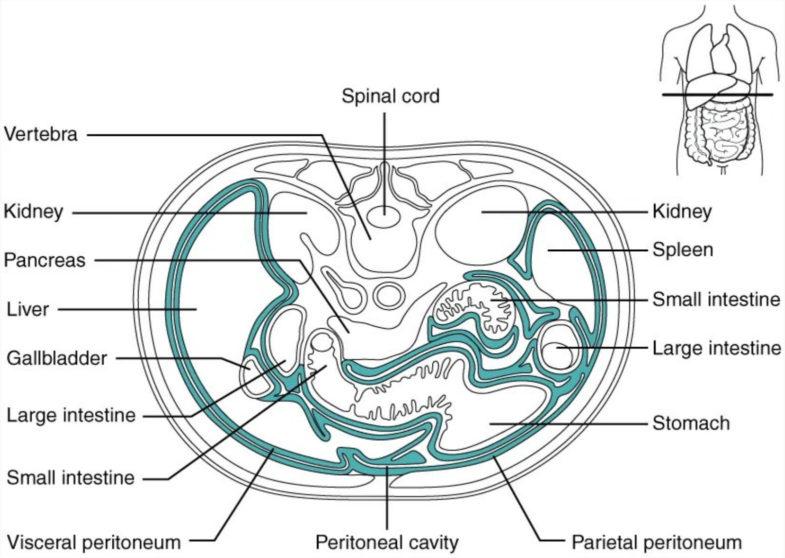
CONCEPT

Two layers:

- Parietal peritoneum: lines the inner body wall
- Visceral peritoneum: covers the organs (= serosa)

Between them: peritoneal cavity with serous fluid (reduces friction)

Intraperitoneal: within the cavity (stomach, jejunum, ileum)
Retroperitoneal: behind the cavity (duodenum, pancreas, kidneys)



Mesenteries: Support, Blood Supply, and Anchoring

CONCEPT

Mesenteries = folds of visceral peritoneum that suspend organs

Carry blood vessels, nerves, and lymphatic vessels to the organs

Two major mesenteries:

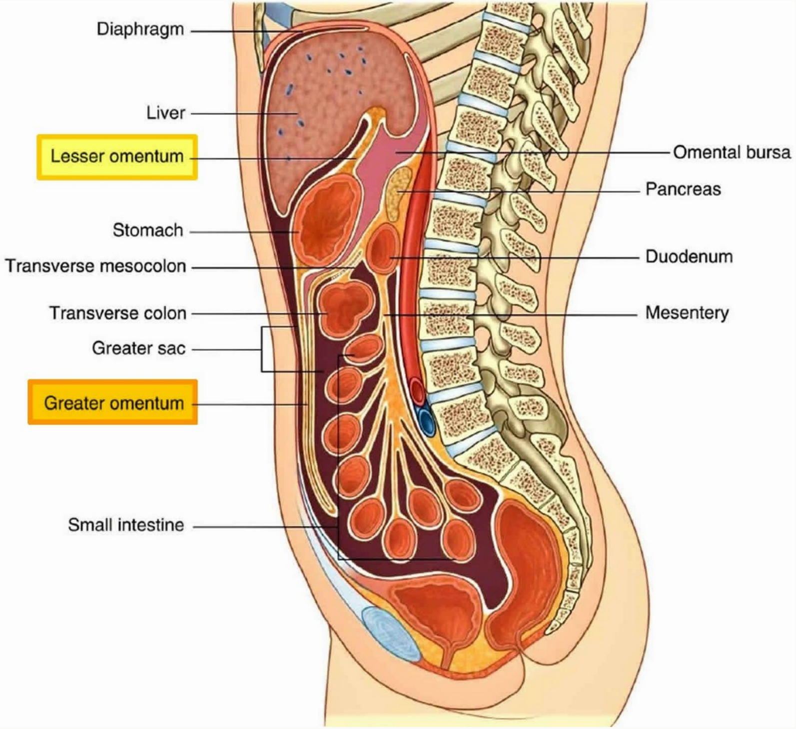
Greater omentum

"Fatty apron" draping from stomach over intestines. Immune function (macrophages).

Lesser omentum

Connects the stomach to the liver.

Mesocolon: mesentery of the large intestine



What's Coming on Wednesday

Wednesday: The journey of food from the mouth through the stomach

- Oral cavity, teeth, salivary glands, swallowing
- Esophagus structure and peristalsis
- The molecular machinery of gastric acid secretion
- GERD: when the system fails

Before Wednesday:

- Start reading OpenStax Ch 23 (or Amerman Ch 22)
- Begin the ALW workbook
- **KC Quiz due Sunday, April 12 (2 attempts, keep highest)**

Your Assignments: Weeks 0 & 1

Assignment	Points	Due
ALW Workbook (Extra Credit)	+3	Sunday, April 12, 11:59 PM
ALW Key Concepts Quiz	24	Sunday, April 12, 11:59 PM
RCS Quiz 1	45.5	Saturday, April 18, 11:59 PM
Application Activity: Ingestion to Absorption	15	Saturday, April 18, 11:59 PM

Start with the ALW and KC Quiz. The RCS and Application Activity will make more sense after all four lectures.

Stay Connected and Get Help

Ask Questions

During class, on Canvas, by email

lagnagiorgio@fhda.edu

Office Hours

Monday & Wednesday, 12:00–12:30 PM, Room 8403

Tutoring & Support

Extra credit opportunities available

Questions?

See you Wednesday at 10:00 AM.